

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 1 Answer Book

NAME: Memo

CLASS: 12 TEACHER:

Question	Mark			Marker	HOD
	H	T	U		
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
TOTAL					
%					

QUESTION 1

	Solution	Marks
1.1.1	$(x-1)(x+2)=4$ $x^2+x-2-4=0$ $x^2+x-6=0 \checkmark$ $(x+3)(x-2)=0$ $x=-3 \checkmark \text{ or } x=2 \checkmark$	(3)
1.1.2	$2x^2 + 5x = 1$ $2x^2 + 5x - 1 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $x = \frac{-(5) \pm \sqrt{(5)^2 - 4(2)(-1)}}{2(2)} \checkmark$ $x \approx 0,19 \checkmark \text{ or } x \approx -2,69 \checkmark$	(3)
1.1.3	$x^2 - 4x - 21 \geq 0$ $(x-7)(x+3) \geq 0 \checkmark$ CV: $x=7$ or $x=-3$ $\checkmark$CV  $\therefore x \leq -3 \checkmark \text{ or } x \geq 7 \checkmark$	(4)

	Solution	Marks
1.1.4	$3^{2x} + 2 \cdot 3^x - 15 = 0$ $(3^x + 5)(3^x - 3) = 0 \checkmark$ $3^x = -5$ or $3^x = 3$ <i>CA both</i> No Solution $\checkmark$ $\therefore x = 1 \checkmark$	(4)
1.1.5	$\sqrt{x-2} + \frac{12}{\sqrt{x-2}} = 7$ Let $\sqrt{x-2} = k$ $\checkmark m$ $k + \frac{12}{k} = 7$ $k^2 - 7k + 12 = 0 \checkmark$ $(k-4)(k-3) = 0$ $k = 4$ or $k = 3$ <i>CA both</i> $\sqrt{x-2} = 4$ $\sqrt{x-2} = 3$ $x-2 = 16$ $x-2 = 9$ $x = 18 \checkmark$ $x = 11 \checkmark$	(5)

	Solution	Marks
1.2	$2x + y = 1$ $x^2 - 3xy + y^2 + 2y = 25 \dots \textcircled{2}$ $y = 1 - 2x \dots \textcircled{1} \checkmark$ Sub ① into ② $x^2 - 3x(1-2x) + (1-2x)^2 + 2(1-2x) = 25 \checkmark^m$ $x^2 - 3x + 6x^2 + (1 - 4x + 4x^2) + 2 - 4x = 25$ $11x^2 - 11x - 22 = 0$ $x^2 - x - 2 = 0 \checkmark$ $(x-2)(x+1) = 0$ $x = 2 \checkmark^{\text{CA both}}$ or $x = -1$ $\therefore y = 1 - 2(2)$ $y = 1 - 2(-1)$ $y = -3 \checkmark^{\text{CA both}}$ $y = 3$	(5)
1.3	$2 \cdot 7^n - 7^{n+1} + 7^{n+2}$ $= 2 \cdot 7^n - 7^n \cdot 7 + 7^n \cdot 7^2$ $= 7^n (2 - 7 + 7^2) \checkmark$ $= 7^n (44) \checkmark$ $= 2(22) 7^n \checkmark$	(3)
		[27]

QUESTION 2

	Solution	Marks
2.1	$5; 7; 9; 11; \dots$ $S_n = \frac{n}{2} [2a + (n-1)d]$ ✓ Correct form $S_{70} = \frac{70}{2} [2(5) + (70-1)(2)]$ ✓ a, d, n $= 5180$ ✓	(3)
2.2	$T_1 \quad ; \quad T_2 \quad ; \quad T_3 ;$ $\quad \quad \quad \swarrow \quad \searrow \quad \quad \swarrow \quad \searrow$ $\quad \quad \quad 5 \quad \quad \quad 7$ $\quad \quad \quad \swarrow \quad \searrow$ $\quad \quad \quad 2$ if $T_3 = 14$; $\therefore T_2 = 7$; $T_1 = 2$ ✓ $2a = 2$ $3(1) + b = 5$ $1 + 2 + c = 2$ $a = 1$ ✓ $b = 2$ ✓ $c = -1$ ✓ $\therefore T_n = n^2 + 2n - 1$ ✓	(5)
2.3	$n^2 + 2n - 1 = 959$ ✓ cA $n^2 + 2n - 960 = 0$ ✓ $(n + 32)(n - 30) = 0$ $n = -32$ or $n = 30$ ✓ N/A ✓	(4)
		[27]

Alternative for 2.2

$$2a = 2$$

$$a = 1 \checkmark$$

$$3a + b = 5$$

$$3(1) + b = 5$$

$$b = 2 \checkmark$$

$$\therefore T_n = an^2 + bn + c$$

$$T_n = n^2 + 2n + c$$

$$\therefore T_3 = 14$$

$$14 = (3)^2 + 2(3) + c \checkmark$$

$$\therefore c = -1 \checkmark$$

$$\therefore T_n = n^2 + 2n - 1 \checkmark$$

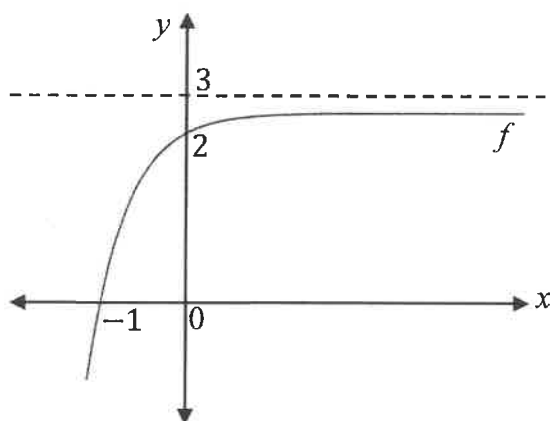
(5)

QUESTION 3

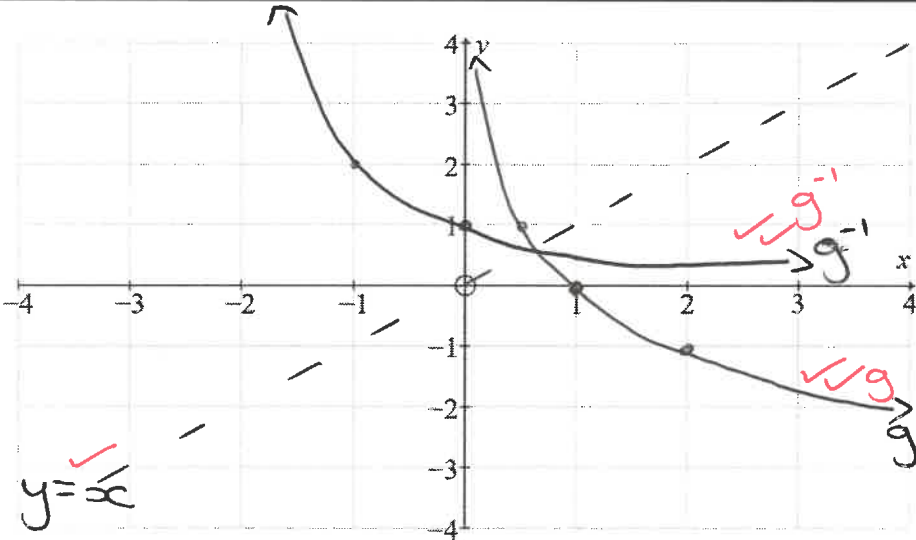
	Solution	Marks
3.1.1	$ar = 6 \dots \textcircled{1} \checkmark$ $ar^6 = 192 \dots \textcircled{2} \checkmark$ $\textcircled{2} \div \textcircled{1}$ $\frac{ar^6}{ar} = \frac{192}{6}$ $r^5 = 32$ $r = 2 \checkmark$ $\therefore a(2) = 6$ $a = 3 \checkmark$	(4)
3.1.2	$S_n = \frac{a(r^n - 1)}{r - 1}$ $\frac{3(2^n - 1) \checkmark}{2 - 1} = 1000\,000 \checkmark$ $2^n - 1 = \frac{1000\,000}{3}$ $2^n = \frac{1000\,000}{3} + 1$ $n = \log_2 \frac{1000\,003}{3} \checkmark \log$ $n \approx 18,35 \checkmark$ $\therefore \text{Minimum } n \text{ is } 19 \checkmark$	(5)

	Solution	Marks
3.2.1	$r = \frac{T_2}{T_1} = \frac{t^2 - 5t + 6}{2t - 4} \checkmark$ $= \frac{(t-3)(t-2)}{2(t-2)}$ $r = \frac{t-3}{2} \checkmark$ Converges: $-1 < r < 1$ $-1 < \frac{t-3}{2} < 1 \checkmark$ $-2 < t-3 < 2$ $1 < t < 5 \checkmark$	(4)
3.2.2	$S_{\infty} = \frac{a}{1-r}$ $0 = \frac{2t-4}{1-\left(\frac{t-3}{2}\right)} \checkmark S_{\infty} \quad \checkmark = 0$ $2t - 4 = 0$ $2t = 4$ $t = 2 \checkmark$	(3)
		[16]

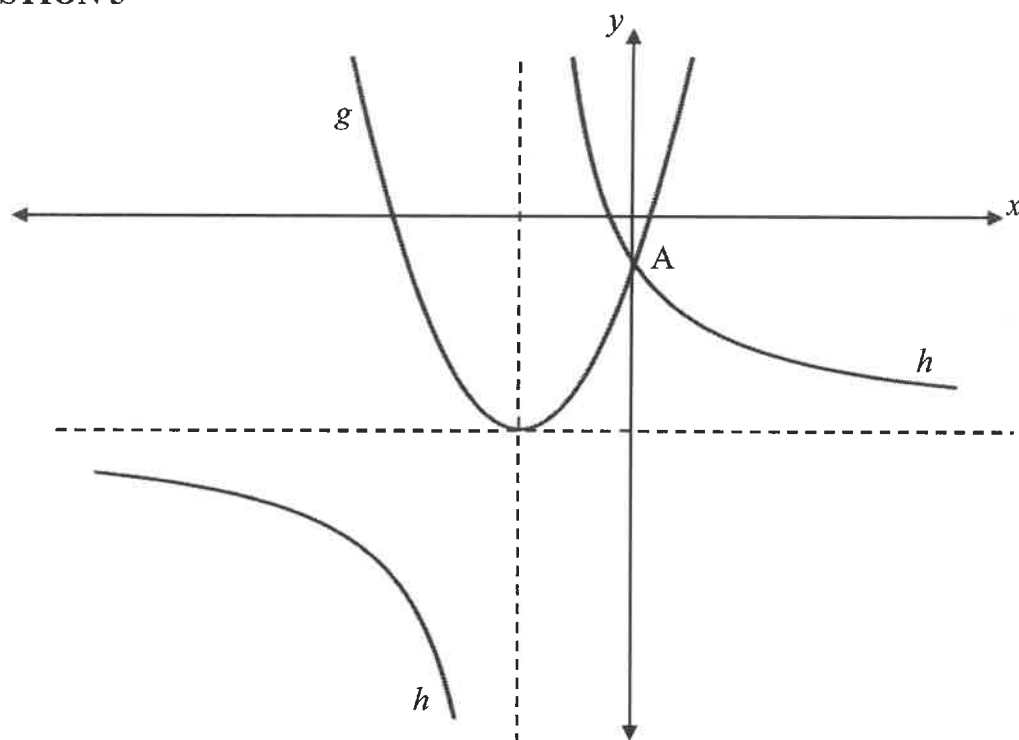
QUESTION 4



	Solution	Marks
4.1.1	$c = 3$ ✓	(1)
4.1.2	$0 = -b^{-1} + 3$ ✓ Subs $(-1; 0)$ $b^{-1} = 3$ $b = \frac{1}{3}$ ✓	(2)
4.1.3	$g(x) = f(x-1) - 4$ ✓ m $= -\left(\frac{1}{3}\right)^{x-1} + 3 - 4$ $= -\left(\frac{1}{3}\right)^x \left(\frac{1}{3}\right)^{-1} - 1$ $= -3\left(\frac{1}{3}\right)^x - 1$ ✓	(3)

	Solution	Marks
4.2.1	$g: y = \log_{\frac{1}{2}} x$ $g^{-1}: x = \log_{\frac{1}{2}} y \quad \checkmark x \leftrightarrow y$ $\therefore y = \left(\frac{1}{2}\right)^x \quad \checkmark$	(2)
4.2.2		(5)
4.2.3	$y > 0 \quad \checkmark$	(1)
		[14]

QUESTION 5



	Solution	Marks
5.1	$x = -1$ ✓	(1)
5.2	$y = \frac{2}{0+1} - 3$ ✓ $x = 0$	(2)
	$y = -1$	
	$A(0; -1)$ ✓ Must be coordinate	
5.3	$y = a(x+1)^2 - 3$ ✓ $p: 2$	(3)
	$-1 = a(0+1)^2 - 3$ ✓ Subs A	
	$a = 2$ ✓	
5.4	$y = -x - 4$ ✓	(2)

	Solution	Marks
5.5	$1 < k < 3$	(2)
5.6	$x < -1$	(2)
5.7	$g(x) = 2(x+1)^2 - 3$ $= 2(x^2 + 2x + 1) - 3$ $g(x) = 2x^2 + 4x - 1$ $\therefore g'(x) = 4x + 4$ $4x + 4 = 2$ $4x = -2$ $x = -\frac{1}{2}$ $\therefore g\left(-\frac{1}{2}\right) = -\frac{5}{2}$ $\therefore m = -\frac{3}{2}$	(6)
	$\therefore g'(x) = 4x + 4$ $4x + 4 = 2$ $4x = -2$ $x = -\frac{1}{2}$ $\therefore g\left(-\frac{1}{2}\right) = -\frac{5}{2}$ $\therefore m = -\frac{3}{2}$	
		[18]

Alternative 5.7

$$g(x) = 2x^2 + 4x - 1$$

$$\text{Point of int: } 2x^2 + 4x - 1 = 2x + m \checkmark$$

$$2x^2 + 2x - 1 - m = 0 \checkmark$$

$$\begin{aligned} \therefore \Delta &= b^2 - 4ac \\ &= (2)^2 - 4(2)(-1-m) \checkmark \\ &= 4 + 8 + 8m \\ &= 8m + 12 \checkmark \end{aligned}$$

$$\text{One root: } \Delta = 0$$

$$8m + 12 = 0 \checkmark$$

$$8m = -12$$

$$m = -\frac{3}{2} \checkmark$$

QUESTION 6

	Solution	Marks
	$f(x) = x^2 - 3$	
	$f'(x) = 2x$ ✓	
	$f^{-1} : x = y^2 - 3$ ✓ $y > 0$	
	$y^2 = x + 3$	
	$y = \pm \sqrt{x+3}$	
	$y = \sqrt{x+3}$ ✓	
	$\therefore 2x = \sqrt{x+3}$ ✓	
	$4x^2 = x + 3$	
	$4x^2 - x - 3 = 0$	
	$(4x+3)(x-1) = 0$	
	$x = -\frac{3}{4}$ or $x = 1$ ✓	
	N/A ✓	
		(6)
		[6]

QUESTION 7

	Solution	Marks
7.1.1	$A = P(1-i)^n$	(2)
	$= 120\,000(1-9\%)^5 \checkmark$	
	$= 74\,883,86 \checkmark$	
7.1.2	$A = P(1+i)^n$	(2)
	$= 120\,000(1+7\%)^5 \checkmark$	
	$= 168\,306,21 \checkmark$	
7.1.3	$F = \frac{x[(1+i)^n - 1]}{i}$	(4)
	$934\,22,35 = \frac{x[(1+\frac{8,5\%}{12})^{61} - 1]}{\frac{8,5\%}{12}} \checkmark \text{ Correct form}$	
	$\checkmark \text{ CA}$	
	$\checkmark n; i$	
	$\therefore x = 1229,73 \checkmark$	

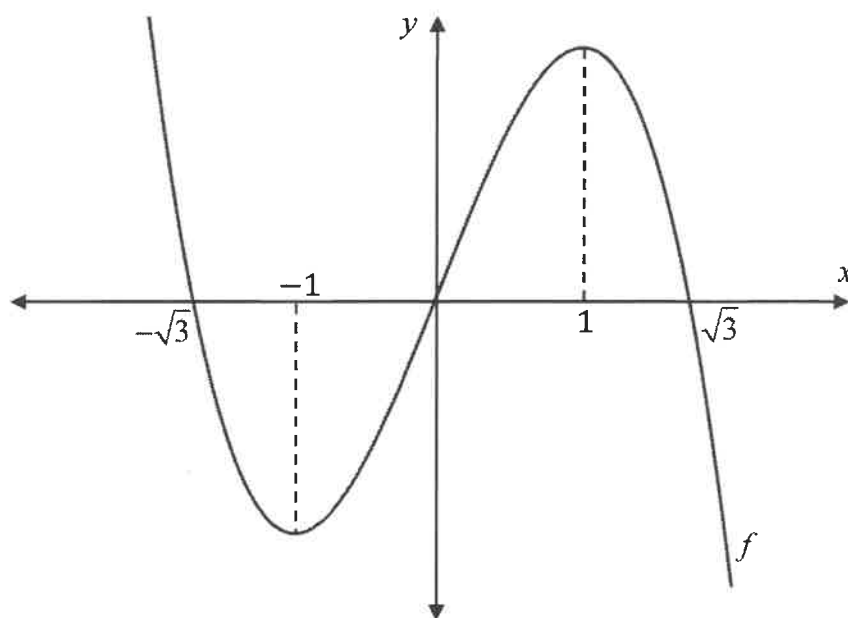
	Solution	Marks
7.2.1	$OB = P(1+i)^n - x \left[(1+i)^n - 1 \right]$ $= 1275000 \left(1 + \frac{9,2\%}{12} \right)^{84} - 11636,02 \left[\left(1 + \frac{9,2\%}{12} \right)^{84} - 1 \right]$ $OB = 1056675,39$	(4)
7.2.2	$\text{Interest} = 1056675,39 \left(1 + \frac{9,2\%}{12} \right)^5$ $= 1097807,15$ $P = x \left[1 - (1+i)^{-n} \right]$ $1097807,15 = x \left[1 - \left(1 + \frac{9,2\%}{12} \right)^{-151} \right]$ $x = 12297,82$	(4)
		[16]

QUESTION 8

	Solution	Marks
8.1	$g(x) = x^3$ $g(x+h) = (x+h)^3$ $= (x+h)(x^2 + 2xh + h^2)$ $= x^3 + 2x^2h + h^2x + x^2h + 2xh^2 + h^3$ $= x^3 + 3x^2h + 3xh^2 + h^3$ $g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \quad \checkmark$ $= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$ $= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h}$ $= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h} \quad \checkmark \text{HCF}$ $= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2$ $= 3x^2 + 3x(0) + (0)^2 \quad \checkmark \text{Apply limit}$ $= 3x^2 \quad \checkmark$	

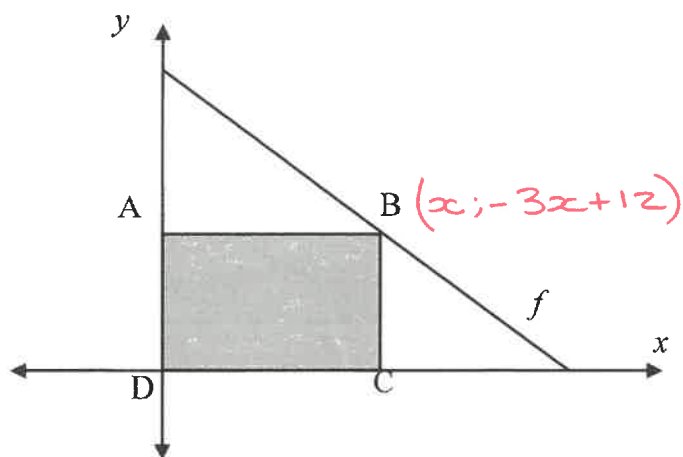
	Solution	Marks
8.2.1	$f(x) = (x-3)(2x+1)$	
	$f(x) = 2x^2 - 5x - 3$ ✓	
	$f'(x) = 4x - 5$ ✓	
		(3)
8.2.2	$\frac{d}{dx} \left(\sqrt[3]{x^5} - \frac{4}{x^4} \right)$	
	$\frac{d}{dx} x^{\frac{5}{3}} - 4x^{-4}$ ✓	
	$= \frac{5}{3} x^{\frac{2}{3}} + 16x^{-5}$ ✓	
		(3)
		[11]

QUESTION 9



	Solution	Marks
9.1	$x \leq -1$ ✓ or $x \geq 1$ ✓ or $x < -1$ or $x > 1$	(2)
9.2	$x = -1$ ✓; $x = 1$ ✓	(2)
9.3	$x < 0$ ✓✓	(2)
9.4	$f(x) = -2x(x + \sqrt{3})(x - \sqrt{3})$ ✓ $= -2x(x^2 - 3)$ ✓ $= -2x^3 + 6x$ ✓	(4)
		[10]

QUESTION 10



	Solution	Marks
10.1	$\text{Area} = AB \times BC \checkmark$ $= x(-3x+12) \checkmark$ $= -3x^2 + 12x$	(2)
10.2	$A = -3x^2 + 12x$ $A' = -6x + 12 \checkmark$ $-6x + 12 = 0 \checkmark$ $-6x = -12$ $x = 2 \checkmark$ $\therefore \text{Max } A = -3(2)^2 + 12(2)$ $= 12 \text{ units}^2 \checkmark$	(4)
		[6]

QUESTION 11

	Solution	Marks
11.1.1	$P(C) \cdot P(D) = P(C \text{ and } D) \checkmark$	
	$P(C) \cdot 0,5 = 0,3$	
	$P(C) = 0,6 \checkmark$	
	$\therefore P(\text{only } C) = 0,6 - 0,3 \checkmark$	
	$= 0,3 \checkmark$	
		(4)
11.1.2	$P(\text{not } C \text{ or not } D) = 1 - 0,3 \checkmark$	
	$= 0,7 \checkmark$	
		(2)

Alternative 10.2

$$A = -3x^2 + 12x$$

$$= -3(x^2 - 4x + (-2)^2 - (-2)^2) \checkmark \checkmark$$

$$= -3[(x-2)^2 - 4]$$

$$= -3(x-2)^2 + 12 \checkmark$$

$\therefore$ Max area $12 \text{ units}^2 \checkmark$

OR

$$A = -3x^2 + 12x$$

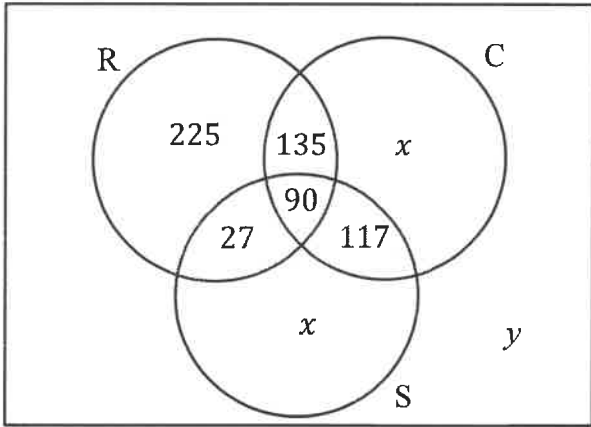
$$x = \frac{-b}{2a}$$

$$= \frac{-(12)}{2(-3)} \checkmark$$

$$= 2 \checkmark$$

$$\therefore \text{max } A = -3(2)^2 + 12(2) \checkmark$$

$$= 12 \text{ units}^2 \checkmark$$

	Solution	Marks
11.2		
11.2.1	$\frac{90}{900} = \frac{1}{10} \quad \checkmark$	(1)
11.2.2	$2x + 225 + 135 + 90 + 27 + 117 = 630 \quad \checkmark$ $2x = 36$ $x = 18 \quad \checkmark$	(2)
11.2.3	$y = 900 - 630 \quad \checkmark$ $y = 270 \quad \checkmark$ $\therefore P(\text{none}) = \frac{270}{900} = \frac{3}{10} \quad \checkmark$	(3)
11.2.4	$P(R \text{ or } C) = \frac{612}{900} \quad \checkmark$ $= 68\% \quad \checkmark$	(2)
		[14]